Financial Institutions and System

Week 14 Guest Lecture: AI/ML in Finance

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Agenda

1. Final Exam Review and Course Wrap-Up
2. Welcome and Session Purpose
3. Guest Speaker Introduction
4. Why This Topic Matters for FIS
5. Lecture: "AI/ML in Finance: Opportunities and Challenges"
6. Q&A and Reflection

Guest Speaker

Cory Baird, Ph.D.

- AI, NLP, ML andForecasting Professional
- Senior Data Scientist, Rakuten
- Visiting Lecturer / Visiting Researcher, University of Tokyo, Center for Advanced Research in Finance (CARF)
- Ph.D. in Public Policy, University of Maryland
- Industry and research experience in:
  - Artificial Intelligence
  - Natural Language Processing
  - Machine Learning
  - Financial Markets
  - Forecasting

- Research interests:
  - Monetary Policy Communication
  - Economic and Financial Shocks
  - Financial Markets
  - Forecasting and Prediction

LinkedIn: https://www.linkedin.com/in/cory-baird-mn/

Why His Voice Matters

Dr. Baird's work sits at the intersection of:

- Artificial Intelligence
- Financial Markets
- Monetary Policy
- Data Science
- Forecasting

His experience spans both academic research and real-world AI applications, providing valuable insight into how machine learning and natural language processing are transforming modern finance.

Selected Research and Professional Interests

Dr. Baird has many ongoing research and applied projects.

You can see the breadth of his active work on his GitHub page.

One project in its final-stage public beta: Monetary Policy Statement Database (MPSD)

- standardized, machine-readable central bank policy-text database
- coverage: 6,693 policy statements from 51 central banks (1990-2024)
- built for research on communication trends, inflation dynamics, and global policy transmission
- current findings include communication trend shifts, cross-country inflation comovement, and sentiment signals linked to global financial conditions
- next expansion adds central bank minutes and broader country coverage

References: Cory Baird GitHub | CentralBankTexts / MPSD

Lecture Focus

AI, NLP, and Machine Learning in Modern Finance

As you listen, think about three questions:

1. Data

How can financial institutions transform large volumes of unstructured text into useful information?

2. Markets

How do markets respond to information released by central banks and policymakers?

3. Prediction

What can machine learning help us forecast better than traditional approaches?

Connecting to This Course

Today's lecture directly relates to several topics covered in Financial Institutions and System:

Financial Markets

- How information affects asset prices

Financial Institutions

- AI adoption in banks and financial firms

Central Banking

- Monetary policy communication

Risk Management

- Forecasting and early-warning systems

Financial Innovation

- Machine learning and NLP applications

Why This Topic Matters

Financial institutions increasingly rely on:

- Artificial Intelligence
- Natural Language Processing
- Machine Learning

to analyze:

- News
- Earnings reports
- Central bank statements
- Social media
- Market sentiment

Understanding these tools is becoming essential for careers in finance, banking, consulting, and policy institutions.

Discussion Prompts for Q&A

AI and Finance

1. Which financial activities are most likely to be transformed by AI over the next decade?

Central Banking

1. Can AI improve our understanding of monetary policy communication?

Financial Markets

1. How quickly do markets incorporate information from policy announcements?

Careers

1. What skills should students develop to work at the intersection of finance and artificial intelligence?

With Respect and Gratitude

Thank you, Dr. Baird, for sharing your expertise and experience with our class.

LinkedIn

Cory Baird LinkedIn Profile

Lecture begins

AI, NLP, Machine Learning, and the Future of Finance

Q&A and Reflection

Thank you for your active participation.